

If you are 19 years old, you have lived about 166,440 hours. If you live to be 80, you will live another 534,360 hours. This test is 1–2 hours of your life. Keep your perspective and don't sweat it. Good luck!

Circle True or False. Read each statement completely before answering. (2 points each)

- | | | | |
|-----|------|-------|--|
| 1. | True | False | Negative feedback most often occurs in a population when N is much larger than K . |
| 2. | True | False | An insect that has evolved to resemble a plant twig will probably be able to avoid parasitism. |
| 3. | True | False | Disease killed off most of the sea urchins around Jamaican coral reefs. The number of species on the coral reef decreased dramatically after the sea urchins died. The sea urchins may have been a keystone species. |
| 4. | True | False | Energy that enters into an ecosystem is eventually lost as metabolic heat. |
| 5. | True | False | Net primary production represents the amount of energy that will be available to tertiary consumers in an ecosystem. |
| 6. | True | False | The carrying capacity of source habitats decreases when the carrying capacity of sink patches increases. |
| 7. | True | False | A population will always grow smaller if immigration is greater than birth rate. |
| 8. | True | False | Any one trophic level supports on average about 10% of the energy of the next lower trophic level. |
| 9. | True | False | The best classification system is that which most closely reflects evolutionary history. |
| 10. | True | False | Mutualism can be thought of as two species that ecologically exploit each other over the long term. |

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect. (2 points each)

11. Consider the food chain grass → grasshopper → mouse → snake → hawk. How much of the chemical energy produced by photosynthesis of the grass (100%) is available to the hawk?
- A. 0.01%
 - B. 10%
 - C. 1%
 - D. 0.1%
 - E. 60%
12. Which of the following areas of study focuses on the exchange of energy and nutrients among individuals of different species?
- A. Behavioral Ecology
 - B. Evolutionary Ecology
 - C. Ecosystem ecology.
 - D. Community Ecology
 - E. Population Ecology
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13. A genetic change that caused a certain *Hox* gene to be expressed along the tip of a vertebrate limb bud instead of farther back helped make possible the evolution of the tetrapod limb. This type of change is an example of
- A. the influence of environment on development.
 - B. heterochrony.
 - C. paedomorphosis.
 - D. a change in a toolkit gene that altered the organization of body parts.
 - E. allometry.
14. A small population of white-footed mice has the same intrinsic rate of increase (r) as a large population. If everything else is equal,
- A. the growth trajectories of the two populations will proceed in opposite directions.
 - B. the two populations will add equal numbers of individuals per unit time.
 - C. the large population will add more individuals per unit time.
 - D. the number of individuals added will depend on the value of N .
 - E. the small population will add more individuals per unit time.
15. Symbiosis is a longer-term ecological interaction
- A. where one species benefits and the other species neither benefits nor is harmed.
 - B. where one species benefits and another species is harmed
 - C. where two or more species partition resources.
 - D. between two or more species.
 - E. where both species benefit.
16. In ecosystems, why is the term cycling used to describe the transfer of chemical elements and nutrients, whereas the term flow is used for the transfer of energy?
- A. Chemical elements are cycled into ecosystems from other ecosystems, but energy constantly flows within the ecosystem.
 - B. Energy flows from one ecosystem to other ecosystems, but chemical elements constantly cycle within an ecosystem.
 - C. Both chemical elements and energy flow within an ecosystem but can cycle to other ecosystems.
 - D. Chemical elements are recycled within ecosystems, but energy flows through and out of ecosystems.
 - E. Both chemical elements and energy are recycled but can flow to other ecosystems.
17. The basic model of metapopulations views the number of suitable habitat patches occupied at a given point in time as a balance between
- A. resource quality and degree of isolation.
 - B. dispersal and birth rate.
 - C. birth rate and local extinction.
 - D. recolonization and local extinctions.
 - E. recolonization and degree of isolation.
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18. A patch in a population has the following four values.

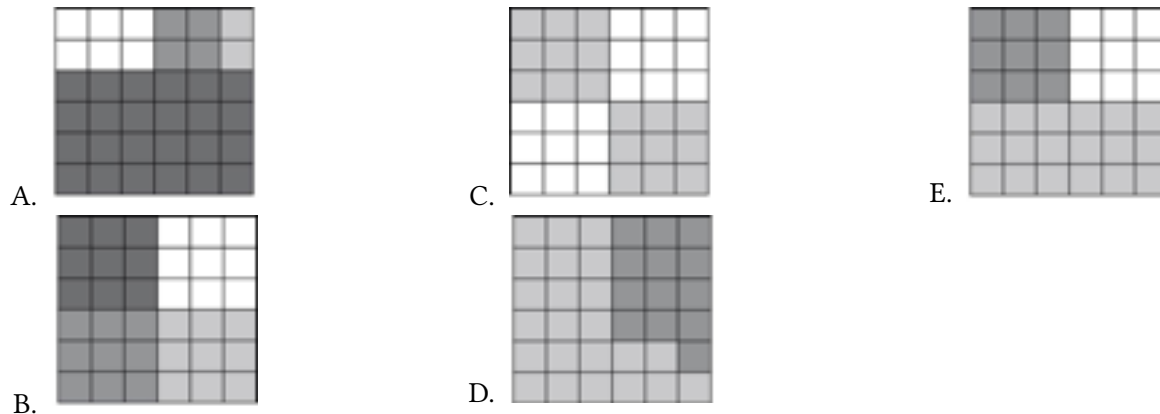
$$b = 0.04 \quad i = 0.06$$

$$d = 0.08 \quad e = 0.04$$

Based only on these values (ignoring habitat quality, patch size, etc.), the patch is most likely to be a

- A. source.
 - B. sink.
19. The attribute that most influence population density and rates of immigration and emigration among habitat patches in a metapopulation are
- A. habitat quality, size of sink patches and isolation of source patches.
 - B. patch isolation, habitat quality, and size of source patches.
 - C. patch size, habitat quality, and isolation of sink patches.
 - D. patch size, patch isolation and habitat quality.
 - E. patch size, patch isolation and species diversity.
20. Assume that you performed an ANOVA. The final probability was $p = 0.035$. Based on this information, which of the following is true?
- A. You would accept the alternative hypothesis that the samples are from the same statistical population.
 - B. You would reject the null hypothesis that the samples are from the same statistical population.
 - C. You would reject the alternative hypothesis that the samples are from different statistical populations.
 - D. You would reject the alternative hypothesis that the samples are from the same statistical population.
 - E. You would accept the null hypothesis that the samples are from different statistical populations.
 - F. You would accept the null hypothesis that the samples are from the same statistical population.
21. A field containing 2000 kg of plant material can support about _____ of tertiary consumers.
- A. 2 kg
 - B. 20 kg
 - C. 10 kg
 - D. 10%
 - E. 90%
 - F. 200 kg
22. The science of naming a new species is called
- A. taxonomy.
 - B. systematics.
 - C. binomial nomenclature.
 - D. phylogeny.
 - E. classification.
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23. A secondary consumer has high abundance. In a trophic cascade, which of the following is probably *not* true.
- A. Actually, all of the above are true.
 - B. The different trophic levels will have alternating high and low abundances.
 - C. The tertiary consumer will have low abundance.
 - D. The primary producer will have low abundance.
 - E. The primary consumer will have low abundance.
24. Which block in the figure below shows the greatest species diversity? (Different shades, including white, represent different species. The small squares indicate relative abundance of individuals present.)



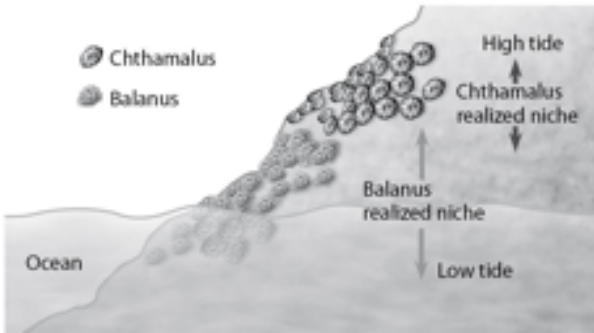
25. Competition for limited resources may influence gene flow by causing
- A. an increased death rate, thereby reducing gene flow.
 - B. some individuals to evolve a new trait to be better competitors.
 - C. some individuals to partition resources, reducing intraspecific competition.
 - D. some individuals to migrate to another population.
 - E. some individuals to move to a different source patch in the same population.
26. The goal of hierarchical classification is to
- A. create monophyletic groups that represent evolutionary history.
 - B. provide a testable hypothesis about organisms with similar structures.
 - C. use a phylogenetic tree to accurately name taxa.
 - D. represent all life on earth with only eight categories, from domain to species.
 - E. provide an easy to memorize arrangement of life on earth.

27. While observing a population of lizards during one week, you notice that the number of adults has increased and is higher at the end of the week than at the start. The number of juveniles is unchanged. The most likely explanation for such an observation would be
- A. decreased emigration.
 - B. decreased death rate.
 - C. decreased immigration.
 - D. increased birth rate.
 - E. increased immigration.
 - F. increased emigration.
28. A change in the body location where two *Hox* genes are expressed caused swimming appendages to evolve into feeding appendages in crabs and other crustaceans. This type of genetic change is an example of
- A. an adaptation to living in a quiet aquatic environment where fewer swimming appendages are needed.
 - B. heterochrony that caused a difference in developmental timing.
 - C. a change in a toolkit gene that altered the arrangement of parts in the body plan.
 - D. paedomorphosis that caused the adult to retain the juvenile feeding appendage.
 - E. allometry that caused the swimming appendages to grow more slowly so they could be used for feeding.
29. Often the growth cycle of one population has an effect on the cycle of another. As moose populations increase, for example, wolf populations also increase. If you consider the logistic equation for the wolf population, which of the factors accounts for the effect of the moose population on the size of the wolf population?
- A. rN
 - B. ΔN
 - C. N
 - D. Δt
 - E. K
 - F. r
30. Given the following values and assuming exponential population growth, what will be the total population size after one generation has elapsed? Round to nearest whole individual.
- $b = 0.09$
 $d = 0.12$
 $N = 18,657$
 $K = 15,000$
- A. 18,097
 - B. 3,657
 - C. 19,217
 - D. 15,000
 - E. 560
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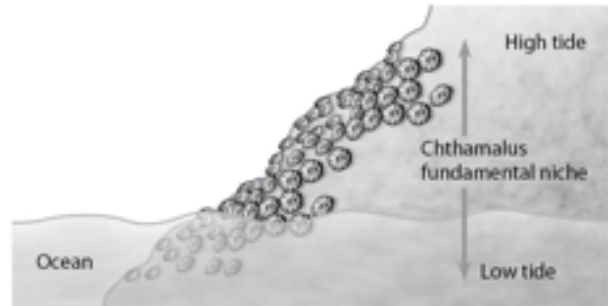
31. Which of the following statements is consistent with the principle of competitive exclusion?
- A. The density of one competing species will have a positive impact on the population growth of the other competing species.
 - B. Evolution tends to increase competition between related species.
 - C. Two species with the same fundamental niche will exclude other competing species.
 - D. Bird species generally do not compete for nesting sites.
 - E. Even a slight reproductive advantage will eventually lead to the elimination of the less well adapted of two competing species.
32. In a mature forest of oak, maple, and hickory trees, a disease causes a reduction in the number of acorns produced by oak trees. Which of the following would *least* likely be a *direct* result of this?
- A. There might be fewer squirrels because they feed on acorns produced by oak trees.
 - B. There might be a decrease in the number of maple seeds as the disease spreads to other trees in the forest.
 - C. There might be fewer mice and seed-eating birds because squirrels that normally eat acorns would eat more seeds and compete with the mice and birds.
 - D. There might be an increase in the number of hickory trees because the competition between hickory nuts and acorns for germination sites would be reduced or eliminated.
 - E. There might be fewer owls because they feed on baby squirrels, mice, and young seed-eating birds, whose populations would be reduced.
33. Which of the following is most likely to contribute to density-dependent regulation of populations?
- A. intraspecific competition for nutrients
 - B. earthquakes
 - C. the removal of toxic waste by decomposers
 - D. fires
 - E. floods
34. White-breasted nuthatches and brown creepers both eat insects that hide in the furrows of bark in hardwood trees. The brown creeper searches for insects by hunting from the bottom of the tree trunk toward the top, whereas the white-breasted nuthatch searches from the top of the trunk down. Their feeding areas rarely overlap on the tree trunk. These hunting behaviors best illustrate which of the following ecological concepts?
- A. competitive exclusion
 - B. resource partitioning
 - C. keystone species
 - D. interspecific competition
 - E. intraspecific competition
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35. In the experiment shown in the figure below, *Balanus* was removed from the habitat shown on the left. Which of the following statements is a valid conclusion of this experiment?

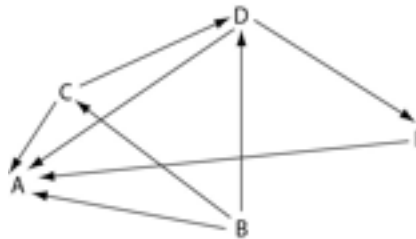
EXPERIMENT Ecologist Joseph Connell studied two barnacle species—*Chthamalus stellatus* and *Balanus balanoides*—that have a stratified distribution on rocks along the coast of Scotland.



RESULTS *Chthamalus* spread into the region formerly occupied by *Balanus*.

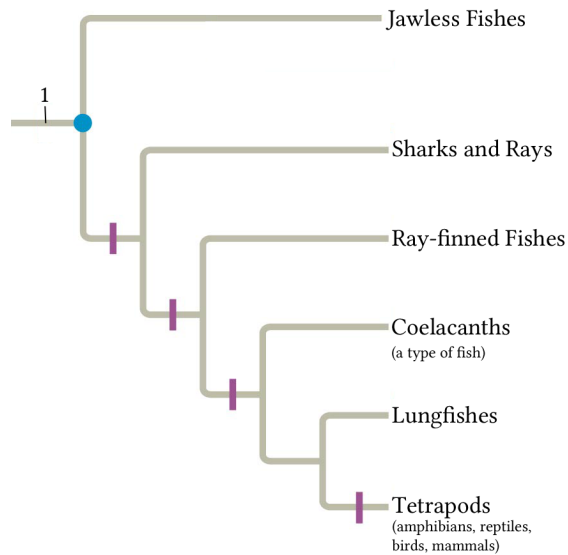


- A. The removal of *Balanus* shows that the realized niche of *Chthamalus* is smaller than its fundamental niche.
- B. These two species of barnacle do not show competitive exclusion.
- C. *Balanus* can survive only in the lower intertidal zone because it is unable to resist dehydration.
- D. *Balanus* is inferior to *Chthamalus* in competing for space on rocks lower in the intertidal zone.
- E. If *Chthamalus* were removed, *Balanus*'s fundamental niche would become larger.
36. Which of the following is the correct and complete arrangement of the hierarchical classification.
- A. Domain, Kingdom, Phylum, Family, Class, Species, Genus.
- B. Domain, Phylum, Family, Class, Order, Genus, Species.
- C. Domain, Kingdom, Phylum, Order, Class, Genus, Family, Species.
- D. Domain, Kingdom, Phylum, Class, Order, Family, Genus, Species.
- E. Domain, Kingdom, Phylum, Order, Class, Family, Genus, Species.
37. Based on the hypothetical food web below, choose one letter that represents an organism that could be both a secondary and tertiary consumer? The arrows point from lower to higher trophic levels. More than one answer is possible. Circle only one answer.



38. (2 points) Write the name of your favorite band or musical artist.

Use the following phylogenetic tree of vertebrate organisms to answer the following questions.



39. Which of the following statements is *not* true?
- A. Jawless fishes are the basal taxon for the entire tree.
 - B. Sharks and rays are a sister group to the group containing ray-finned fishes, coelacanths, lungfishes, and humans.
 - C. Coelacanths are more closely related to lungfishes than tetrapods.
 - D. Ray-finned fishes are more closely related to coelacanths than sharks.
40. The science of ichthyology is the study of fishes, a group that includes all groups shown except for tetrapods. Therefore, ichthyology studies
- A. a clade because it includes all of the descendants but not their common ancestor.
 - B. a monophyletic group because it includes all of the the descendants but not their common ancestor.
 - C. a monophyletic group because it includes all fish descendants shared with jawless fishes.
 - D. a non-monophyletic group because it includes all of the the descendants but not their common ancestor.
 - E. a non-monophyletic group because the group does not include all descendants of their common ancestor.
 - F. a clade because the group of fishes includes all fish-like organisms that swim in water.

The last 10 questions are on the next page. Deep breath and onward!

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice. (2 points each)

Aposomatic Coloration	Gross Primary Production	Primary Producers
Clade	Interspecific Competition	Realized Niche
Classification	Intraspecific Competition	Resource Partitioning
Commensalism	Keystone Species	Richness
Competitive Exclusion	Logistic Growth	Secondary Consumers
Cryptic coloration	Mimicry	Symbiosis
Cryptic species	Mutualism	Taxon
Diversity	Net Primary Production	Taxonomy
Exponential Growth	Parasitism	Tertiary Consumers
Fundamental Niche	Primary Consumers	Trophic Cascade

41. _____ The total number of species present in the community.
42. _____ Occurs when a population has a carrying capacity.
43. _____ A tasty species looks like a poisonous species.
44. _____ Another term for a monophyletic group.
45. _____ A predator species that increases community diversity.
46. _____ Occurs when individuals of different species require the same limited resource.
47. _____ A poisonous species warns a potential predator.
48. _____ Based on the resources actually used by a species.
49. _____ Species divide shared resource, increasing diversity.
50. _____ Total amount of sun's energy converted to organic matter.

Music in the soul can be heard by the universe. Keep a song in your heart, do good things, and have a great break!
